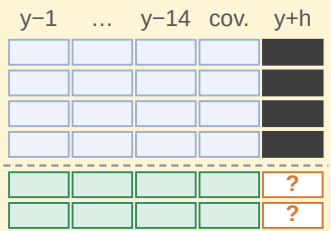
How the boosted trees make a forecast

LightGBM, XGBoost and CatBoost on lagged values · supervised baselines trained on each dataset · Darts

1. Lags become columns

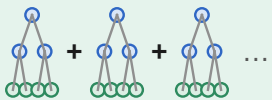
Each forecast origin becomes one row of a table: recent values of the target and covariates.



Daily: the last 14 days (temperature also every 3rd day of the same season a year ago), past-only covariates over 7 days, known-ahead ones every 10th future day. Monthly: 12 and 6 months.

2. Trees, one after another

Boosting adds small decision trees one at a time; each new tree fits what the earlier ones still get wrong.



each tree fixes what is left

100 trees; tree size and learning rate tuned.

3. Every future step at once

The forecast is direct: no step is fed back. The three libraries share out the horizon differently.

LightGBM: one model per future step and series

XGBoost: one booster; each round adds a tree for every step and series

CatBoost: one ensemble predicts every step and series together (MultiRMSE)

4. Train once, reuse

Trained once per dataset on the rows before the first test window (daily: the last 4,096), then every window is forecast from its own origin with no retraining.

point forecast

Optuna tries a few settings on validation windows inside the training rows; the first is the default.

Hand-picked lags and trees: the classic machine-learning answer to forecasting.

Prices as log differences · min-max scaled on the training rows · direct forecast of 90 days or 24 months · 100 trees